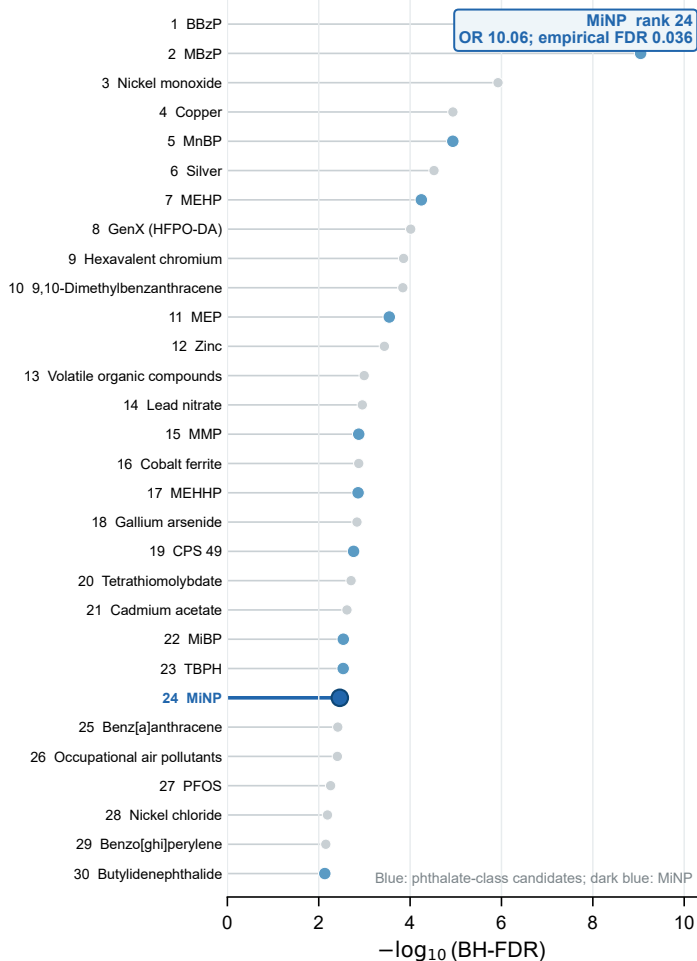
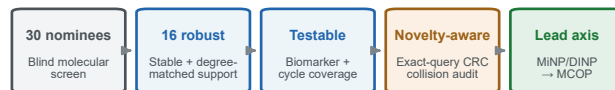


Multistage prioritization advances a rank-24 MiNP signal to a human-testable DINP exposure axis

A Hypothesis-agnostic Top-30 nominees



B Progressive triage prioritizes actionability



Rank alone did not determine advancement

C Audited evidence matrix for representative candidates

	Molecular stability	Degree support	Human biomarker	NHANES cycles	CRC PubMed hits*	Human CRC gate
1 BBzP	Yes	Yes	MBzP proxy	10	0	Null†
2 MBzP	Yes	Yes	MBzP	10	0	Null
3 Nickel monoxide	Yes	Yes	—	—	3	NE
7 MEHP	Yes	Yes	MEHP	10	4	Null
8 GenX	Yes	Yes	—	—	0	NE
18 Gallium arsenide	Yes	Yes	—	—	0	NE
24 MiNP/DiNP	Yes	Yes	MCOP	7	0	Positive

*Exact Title/Abstract query audit (23 Aug 2026); not an exhaustive review. †BBzP represented by urinary MBzP.

D DINP-axis biomarker translation

